

Varsha M - Data analysis - Task 1

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

```
df=pd.read_csv("online_retail.csv")
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 82485 entries, 0 to 82484
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0   InvoiceNo    82485 non-null  object
1   StockCode   82485 non-null  object
2   Description  82230 non-null  object
3   Quantity    82485 non-null  int64
4   InvoiceDate  82485 non-null  object
5   UnitPrice   82485 non-null  float64
6   CustomerID  52406 non-null  float64
7   Country     82485 non-null  object
dtypes: float64(2), int64(1), object(5)
memory usage: 5.0+ MB
```

```
df.describe()
```

	Quantity	UnitPrice	CustomerID
count	82485.000000	82485.000000	52406.000000
mean	8.515451	5.495747	15346.807675
std	369.691978	130.137907	1769.642205
min	-74215.000000	0.000000	12346.000000
25%	1.000000	1.250000	13862.000000
50%	3.000000	2.510000	15311.000000
75%	8.000000	4.240000	17001.000000
max	74215.000000	16888.020000	18283.000000

```
df.isnull().sum()
```

```
0
InvoiceNo    0
StockCode    0
Description  255
Quantity     0
InvoiceDate  0
UnitPrice    0
CustomerID   30079
Country      0
```

dtype: int64

```
df=df.dropna(subset=['CustomerID'])
```

```
df['Description']=df['Description'].fillna('Unknown')
```

```
df.isnull().sum()
```

```
0
InvoiceNo    0
StockCode    0
Description   0
Quantity     0
InvoiceDate  0
UnitPrice    0
CustomerID   0
Country      0
```

dtype: int64

```
df['InvoiceDate']=pd.to_datetime(df['InvoiceDate'])
df['TotalPrice']=df['Quantity']*df['UnitPrice']
```

```
avg_purchase=df['TotalPrice'].mean() ##Descriptive Statistics
print("Average Purchase Amount:",avg_purchase)
```

Average Purchase Amount: 21.135370949891232

```
purchase_freq=df.groupby('CustomerID')['InvoiceNo'].nunique().mean()
print("Average Purchase Frequency:",purchase_freq)
```

Average Purchase Frequency: 2.17032967032967

```
snapshot_date=df['InvoiceDate'].max() ##RFM Feature Engineering
rfm=df.groupby('CustomerID').agg({
    'InvoiceDate':lambda x:(snapshot_date-x.max()).days,
    'InvoiceNo':lambda x:len(x),
    'TotalPrice':lambda x:x.sum()})
rfm.columns=['Recency','Frequency','Monetary']
rfm.head()
```

	Recency	Frequency	Monetary
CustomerID			
12346.0	17	2	0.00
12347.0	8	60	1187.18
12348.0	10	23	1120.24
12350.0	1	17	334.40
12356.0	17	36	2271.62

```

scaler=StandardScaler() ## normalization
rfm_scaled=scaler.fit_transform(rfm)
rfm_scaled=pd.DataFrame(rfm_scaled,columns=rfm.columns,index=rfm.index)
rfm_scaled.head()

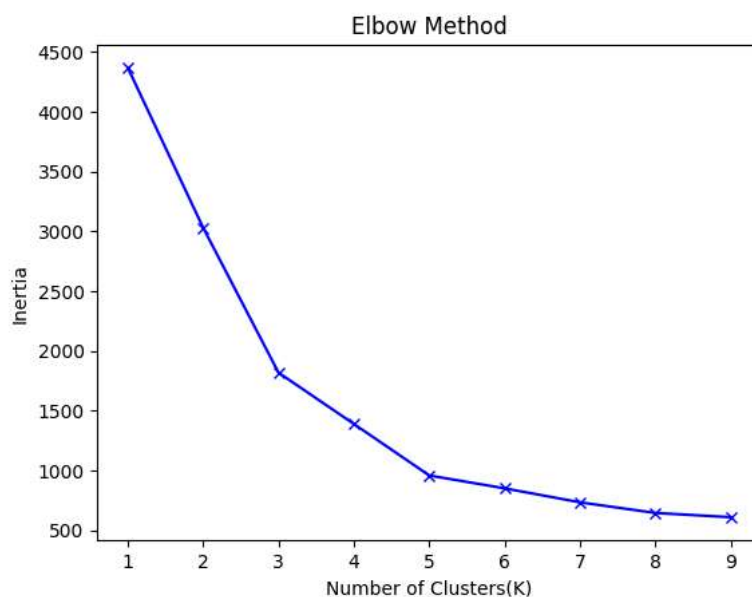
```

	Recency	Frequency	Monetary
CustomerID			
12346.0	-0.535193	-0.671612	-0.384970
12347.0	-0.937878	0.474310	0.215808
12348.0	-0.848393	-0.256709	0.181932
12350.0	-1.251078	-0.375252	-0.215745
12356.0	-0.535193	0.000136	0.764593

```

inertia=[]          ## elbow method
K_range=range(1,10)
for k in K_range:
    kmeans=KMeans(n_clusters=k,random_state=42)
    kmeans.fit(rfm_scaled)
    inertia.append(kmeans.inertia_)
plt.figure()
plt.plot(K_range,inertia,'bx-')
plt.xlabel('Number of Clusters(K)')
plt.ylabel('Inertia')
plt.title('Elbow Method')
plt.show()

```



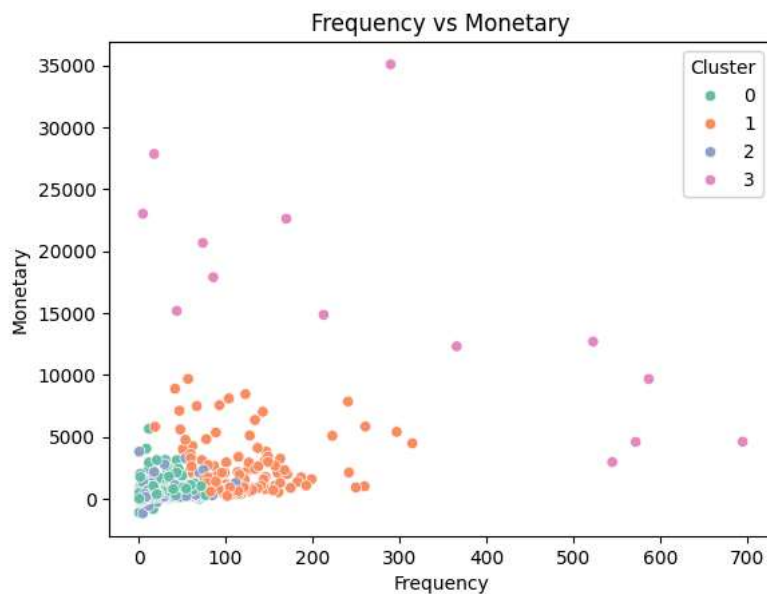
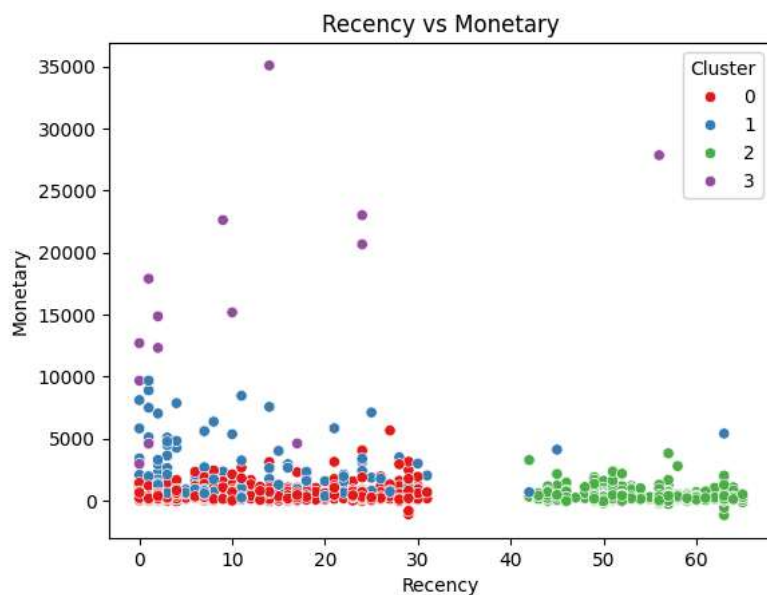
```

kmeans = KMeans(n_clusters=4, random_state=42)
rfm['Cluster'] = kmeans.fit_predict(rfm_scaled)
rfm.head()

```

CustomerID	Recency	Frequency	Monetary	Cluster
12346.0	17	2	0.00	0
12347.0	8	60	1187.18	0
12348.0	10	23	1120.24	0
12350.0	1	17	334.40	0
12356.0	17	36	2271.62	0

```
plt.figure()
sns.scatterplot(x=rfm['Recency'],y=rfm['Monetary'],hue=rfm['Cluster'],palette='Set1')
plt.title("Recency vs Monetary")
plt.show()
plt.figure()
sns.scatterplot(x=rfm['Frequency'],y=rfm['Monetary'],hue=rfm['Cluster'],palette='Set2')
plt.title("Frequency vs Monetary")
plt.show()
```

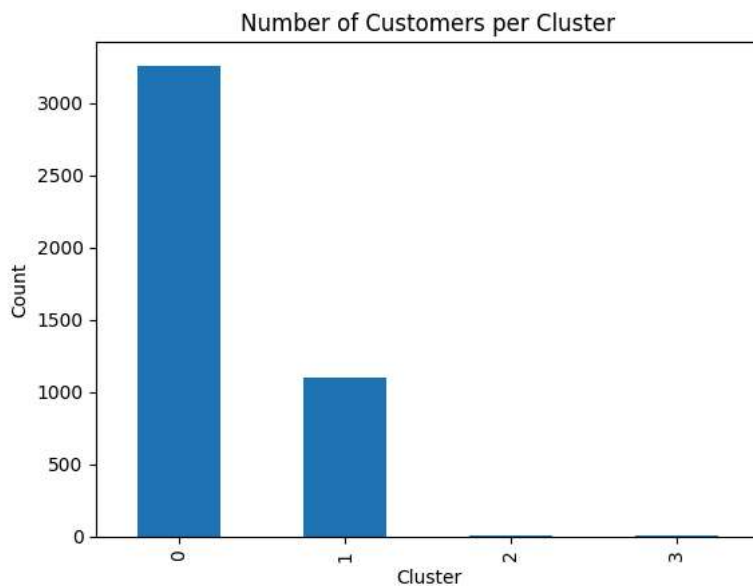


```
cluster_profile = rfm.groupby('Cluster').mean()
print(cluster_profile)
```

	Recency	Frequency	Monetary
Cluster			

0	12.851902	24.982337	507.101155
1	12.483221	113.986577	2154.456107
2	55.096948	23.064632	340.355907
3	11.428571	299.142857	15985.828571

```
plt.figure()
rfm['Cluster'].value_counts().plot(kind='bar')
plt.title("Number of Customers per Cluster")
plt.xlabel("Cluster")
plt.ylabel("Count")
plt.show()
```



```
## Customer per cluster visual
import matplotlib.pyplot as plt
import seaborn as sns
cluster_counts = pd.Series(kmeans.labels_).value_counts().sort_index()
plt.figure(figsize=(7, 5))
sns.barplot(x=cluster_counts.index, y=cluster_counts.values, hue=cluster_counts.index, palette="Blues", legend=False)
plt.title("Number of Customers per Cluster")
plt.xlabel("Cluster")
plt.ylabel("Number of Customers")
plt.show()
```

